

Muhammad Fayaz Khan

LLM Engineer | RAG Systems • Agentic AI • FastAPI • LangChain
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PROFESSIONAL SUMMARY

LLM Engineer specializing in Enterprise-grade RAG Systems and Agentic AI workflows. Experienced in building production-ready AI applications using FastAPI, LangGraph, LangChain, and Vector Databases with a focus on retrieval accuracy, latency optimization, and LLM cost reduction. Proven ability to design end-to-end AI pipelines, from data ingestion and chunking strategy to deployment and evaluation, with measurable, demoable results.

TECHNICAL SKILLS

Generative AI & RAG	RAG Pipeline, Hybrid Search (Dense + BM25), Neural Reranking, Self-Querying, RAGAS Evaluation
Agentic AI	LangGraph, CrewAI, LangChain, Tool Calling, Multi-Agent Workflows, MCP
Backend & APIs	FastAPI, Python (Async/Await, Decorators, Generators), Pydantic v2, Instructor Library
Vector Databases	ChromaDB, Qdrant, Pinecone
Core AI / ML	Machine Learning, Deep Learning (PyTorch), Fine-Tuning (SFT, DPO, LoRA), CNNs, Scikit-Learn
DevOps & Tools	Docker, Docker Compose, Git/GitHub, Streamlit, AWS
LLM Providers	OpenAI, Anthropic Claude, Google Gemini, Groq

PROJECTS PORTFOLIO

CompetiTrack AI — Competitor Price & Stock Monitoring Agent [LangGraph · Playwright · FastAPI · Streamlit · Groq]
Autonomous agent that scrapes competitor e-commerce sites, detects price/stock changes via a branching LangGraph workflow, and sends automated alerts only on real changes. Deployed with a FastAPI backend and Streamlit dashboard.

UrduGPT — Urdu LLM Fine-Tuning Pipeline [PyTorch · Transformers · LoRA · SFT · DPO · TRL]
End-to-end LLM engineering project for Urdu language modeling using Supervised Fine-Tuning (SFT) and Direct Preference Optimization (DPO) to align model outputs with human preferences.

BizBrain — Business PDF Agent [FastAPI · ChromaDB · RAGAS · LangChain · Pydantic · Chainlit · Docker]
Enterprise FAQ agent that ingests business PDFs and answers domain-specific queries using Hybrid Search and Cross-Encoder Reranking, with persistent chat history for real client deployment.

Enterprise RAG Platform [FastAPI · ChromaDB · RAGAS · LangChain]
Production-grade RAG system using Recursive Semantic Chunking and Cross-Encoder Reranking for querying private PDF data, achieving high faithfulness and answer relevancy via RAGAS evaluation.

Multi-Functional AI Agent [LangChain · Pydantic v2 · Tool Calling]
Autonomous agent with structured tool calling and JSON outputs to handle complex, multi-step business workflows.

Rossmann Sales Predictor / Churn Predictor [Python · Scikit-Learn · Feature Engineering]
Time-series retail sales forecasting model and a customer churn classification model, both with advanced feature engineering, ensemble methods, and precision/recall evaluation.

KEY COMPETENCIES

- Optimizing LLM API costs and latency; implementing LLM caching (GPTCache) to cut API costs by up to 40%
- Building stateful, cyclical multi-agent workflows using LangGraph
- Designing RAG 2.0 pipelines with Hybrid Search and Neural Reranking for enterprise clients
- Fine-tuning LLMs with SFT and DPO for low-resource languages
- Building automated web scraping and monitoring pipelines with Playwright and scheduled triggers
- Red-teaming and evaluating LLM outputs with RAGAS and LLM-as-Judge frameworks

EDUCATION

12th Grade (Intermediate, Computer Science) — Completed
Self-Taught LLM Engineering Roadmap (2026)